

Exercise 5 — Safety and Interlock Logic

Programmable Logic Controllers

Lecture: Real-Time Systems — Summer Semester 2026

Programming language: Ladder Diagram (LD)

Scenario

You are programming the safety control system for an industrial hydraulic press used in metal stamping. The press operates in two distinct modes — Setup mode and Production mode — each with different safety rules. In Setup mode a technician may open guards to change tooling or perform maintenance. The ram is allowed to move only at reduced speed and only with a supervisor key switch inserted. In Production mode all guards must be closed, full press force is used, and two-hand control is fully enforced.

The hydraulic system requires time to build sufficient pressure before any ram operation is permitted. The press also has jam detection — if the ram does not complete its stroke within 4 seconds, a jam fault is raised automatically. Read the entire specification before writing any logic. All requirements interact with each other.

Variable table

Variable	Address	Type	Convention	Description
Left_Hand	%QX0.0	Input (BOOL)	NO	Operator left hand control
Right_Hand	%QX0.1	Input (BOOL)	NO	Operator right hand control
Guard_Front	%QX0.2	Input (BOOL)	NC	TRUE when front guard is OPEN
Guard_Left	%QX0.3	Input (BOOL)	NC	TRUE when left guard is OPEN
Guard_Right	%QX0.4	Input (BOOL)	NC	TRUE when right guard is OPEN
E_Stop	%QX0.5	Input (BOOL)	NC	TRUE when released, FALSE when pressed
Mode_Switch	%QX1.0	Input (BOOL)	NO	FALSE = Setup mode, TRUE = Production mode
Key_Switch	%QX1.1	Input (BOOL)	NO	TRUE when supervisor key is inserted
Stroke_Sensor	%QX1.2	Input (BOOL)	NO	TRUE when ram has reached full stroke position
Pressure_Pump	%QX1.3	Output (BOOL)	—	TRUE to run hydraulic pump
Press_Ram_Full	%QX1.4	Output (BOOL)	—	TRUE for full-speed production stroke
Press_Ram_Slow	%QX1.5	Output (BOOL)	—	TRUE for reduced-speed setup stroke
Safety_OK	%QX1.6	Output (BOOL)	—	TRUE when system is ready for press operation
Fault_Light	%QX1.7	Output (BOOL)	—	Latches TRUE on any fault condition
Pump_Run_Light	%QX2.0	Output (BOOL)	—	TRUE when pump is running
Jam_Fault_Light	%QX2.1	Output (BOOL)	—	TRUE when ram jam is detected

Variable table notice

All variables listed above are MUST-HAVE. Their names and addresses cannot be changed.

All variable types are BOOL.

In OpenPLC, all variable classes must be set to Local due to validation requirements.

You may declare additional internal variables (timers, memory bits at %MX addresses) as needed.

Background — permissive logic

A permissive chain means an output is only allowed when ALL safety conditions are simultaneously TRUE. In ladder this is a series chain of contacts — one open contact anywhere blocks the output. This is the primary safety mechanism for the ram outputs.

Background — NC sensor contacts in ladder

Guard sensors are NC (Normally Closed). They output TRUE when the guard is OPEN. To express that a guard is closed in ladder you need a NC contact on the guard variable — a NC contact passes current when the variable is FALSE. E_Stop is also NC — it outputs TRUE when released and FALSE when pressed.

Background — timers used in this exercise

A TON (on-delay) timer starts counting when its input is TRUE and activates its Q output after the preset duration. It is used here for pressure stabilization (5 seconds) and jam detection (4 seconds).

System specification

Hydraulic pump control

The pump runs whenever the E-stop is released and no fault is latched. It does not depend on mode or guard state. A Pump_Run_Light mirrors the pump state exactly. When the pump starts, a 5 second pressure stabilization period begins. During these 5 seconds no ram operation is permitted. Only after stabilization completes may the ram operate.

Production mode — Press_Ram_Full

Mode_Switch must be TRUE. All of the following must be TRUE simultaneously for Press_Ram_Full to activate: E-stop released, all three guards closed, stabilization complete, both hand buttons pressed, pump running, no fault latched, no jam fault latched, and Press_Ram_Slow FALSE. If any condition becomes FALSE during a stroke, Press_Ram_Full must deactivate immediately. Press_Ram_Full must never activate in Setup mode under any conditions.

Setup mode — Press_Ram_Slow

Mode_Switch must be FALSE. The required conditions are: E-stop released, Key_Switch TRUE, stabilization complete, both hands pressed, pump running, no fault, no jam fault, and Press_Ram_Full FALSE. Guards may be open in Setup mode — guard state is not a condition for the slow ram. Press_Ram_Slow must never activate in Production mode.

Direction interlock

Press_Ram_Full and Press_Ram_Slow must never be TRUE simultaneously. Each output has the other's negation as a condition in its permissive chain.

Fault light

The fault light latches on when any of the following occur: E_Stop is pressed at any time; any guard opens while in Production mode and the pump is running; or the Mode_Switch changes position in either direction. The fault light clears only when E_Stop is released AND both hand buttons are pressed AND all guards are closed.

Jam detection

A TON timer monitors the ram stroke. If the ram has been active for 4 seconds without the stroke sensor triggering, a jam is confirmed. On confirmation both ram outputs must deactivate and Jam_Fault_Light must latch on. The jam clears when both hands are released with E_Stop released.

Safety OK light

The Safety_OK light is mode-dependent. In Production mode it requires: E_Stop released, all guards closed, stabilization complete, pump running, and no fault latched. In Setup mode it requires: E_Stop released, Key_Switch TRUE, stabilization complete, pump running, and no fault latched.